Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

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M1

Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I

1 Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I

Study focus: Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000218 · EV0000219

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module 6 on corrosion chemistry, and today's class we are going to talk about corrosion control.

Key details

- In that we will look at metallic coating and specifically galvanization and tinning.
- When we talk of corrosion control, the first thing which comes to our mind is to create a barrier between the surface and the environment.
- So protective coating is a very good method by which we can control corrosion to a great extent.
- In protective coating, we have metallic coating, inorganic coating, and organic coating.
- Then another method of corrosion control is to use corrosion inhibitors.
- Again, we have two classes here: anodic inhibitors and catodic inhibitors.
- The other method would be cathodic protection, and in that we are going to study about two methods in there, that is sacrificial anode method and the impressed cathodic current method.
- Finally, we will study about anodic protection, which is also known as impressed anodic current method.
- The first type of coating we are going to look at is metallic coating.
- Corrosion of the base metal can be prevented by coating it with a layer of another metal over it.
- Therefore, we will have two types of coatings which are possible.

- One is called anodic metal coating where we are going to coat a layer of metal which is anodic with respect to the base metal.
- Which is cathodic with respect to the base metal.
- As I said just now, the coating metal is more active with respect to the base metal.
- Is not completely covered, it is not going to undergo pitting corrosion.
- This is a very big advantage of anodic metal coating.
- With another metal, say this is M2.
- To the base metal.
- Therefore, even if there is a gap here, we will find that if the coating is not completely covering the surface, there is a gap on the surface.
- You have studied that.
- This will not give rise to pitting corrosion because as long as the anode is in contact with the.
- Cathode, it is the anode which undergoes corrosion.
- That is the biggest advantage of having anodic metal coating.
- When we cover the metal with zinc on the surface, like zinc is acting as the coating, then this process is known as galvanization.
- It's usually on iron.
- Coating a layer of zinc on iron would be referred to as galvanization.
- What would be the various steps involved in this process?
- Make sure that the surface is absolutely clean because we know any coating when it is made on a surface, the smoother the surface, the better the coating is going to adhere to the surface.
- So the first and foremost treatment before.
- coating any surface is to make it absolutely clean.
- So in that the first step would be.

Example or application from the lecture

- Let us start with protective coatings.
- A good example would be if you have an iron article and you are covering it with zinc metal, this would be anodic metal coating and then there is another method where we which we call it as the cathodic metal coating where we are going to coat the metal with another metal.
- And an example for that would be an iron article which is coated with tin.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 05:00–09:59

The iron sheet is passed to an organic solvent or a caustic solution to remove any oil or grease which may be present on the surface.

Key details

- Next, it has to be washed with dilute sulfuric acid.
- This process is known as pickling.

- The iron sheet is passed through dilute sulfuric acid so that.
- Any rust, steel, scale, or dust can be removed from the surface.
- And now the next treatment is again very important.
- It is going to pass through an aqueous solution of zinc chloride and ammonium chloride.
- What happens when we use this flux onto the surface is that it will prevent the oxidation.
- Once that is done, it is dried.
- And finally, we come to the main tank, that is the galvanizing tank, where we are going to dip the article into a zinc into molten zinc.
- Of course, if you want a coating of zinc onto the surface, it should be molten zinc, and that melting point of zinc is around 425, 430.
- And finally, excess zinc which is present on the iron sheet has to be removed by passing it through hot rollers.
- So this would be the surfa the method of galvanization, the whole process of galvanization.
- So as I told you, first we need to prepare the surface before it is galvanized.
- in with dilute estrus support again there is rinsing and then finally the flux and then it is dried and you have then it gets into the zinc bar.
- Now, this is called dry method.
- There's also a wet method where the flux need not be separate.
- What is done is the zinc bath itself is covered with this flux.
- So it's a wet method.
- That is also done in some cases.
- Galvanized iron,, we have heard of GI pipes and GI sheets.
- They are called galvanized iron sheets and pipes, which are which is used in building construction material.
- So, as I mentioned just in the beginning of the class, the advantage of this galvanization is that even if the zinc coat peels off or there are gaps on the surface of the gaps in the coating we find that the base metal iron will never get corroded because as long as zinc is present in contact with the iron it is zinc which gets corroded and not the iron article.
- Which is underneath.
- But there is also a serious disadvantage.
- What is the disadvantage?
- That galvanized articles are not used for preparing and storing food because the zinc which is present on the surface is going to dissolve in dilute acids, and that is going to produce.
- So therefore we find that the galvanized article the galvanized sheets cannot be used for storing food.
- making those storing food storing containers.
- So otherwise galvanization is very good because as long as zinc is there, we don't have any problem.
- So that means we need to look for another type of coating, and that is where we talk about the.
- The coating metal is less active with respect to the base metal.
- Means it is more noble as compared to the base metal.
- Now because it is more noble, as long as it covers the entire surface, corrosion will be much less.

Example or application from the lecture

- Let us see this picture and understand.

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Core explanation · 10:00–14:59

But there is a serious problem here, which we have talked about many times.

Key details

- If the coating is not uniform.
- Metal, if there are gaps then now what has happened because the coating is cathodic with respect to the base metal, this gap is going to expose the metal M1, which is now acting as the anode.
- So you have a case of small anode and large cathode, just like we understood in during pitting correct studying pitting corrosion, when we were studying pitting corrosion, and therefore this is going to cause severe fitting corrosion and the metal actually underneath is going to get more damaged than if it was left uncovered.
- How would we define tinning?
- Coating a layer of tin over iron will be called the tinning process.
- Again we understand the article.
- Has to be dipped.
- Now this method is known as the hot dipping method.
- In fact, even the galvanization which we did just now is known as the hot dipping method.
- It's understood why because we are dipping the article in the molten zinc which is hot.
- So therefore we need to dip the article in molten tin, but before that again we need to.
- Take care, I mean we need to do the surface preparation, and here specifically, because we know if there are gaps left on the surface, it is going to result in intense corrosion or pitting corrosion of the article which we want to protect.
- Therefore, we need to take special precautions.
- First the steps are same.
- We are going to pass it through organic solvent or caustic solution to remove the oil or grease present on the surface, and then it is washed with dilute H2O4.
- This process is called pickling to remove any rust, scale, dust, etc.
- And then next step is very important where it is going to be treated with zinc chloride flux.
- It is going to make sure that the tin coating adheres to the surface.
- And then it is dipped in the in molten tin.
- Centigrade, so that it's in the molten form, and then your article is dipped in it.
- Obviously, it is going to get a tin coating.
- Excess tin now is removed.
- By passing through hot rollers, and now it is then immersed in palm oil.
- It prevents the surface oxidation of the freshly coated tin.
- Here we have not shown the organic solvent or the caustic solution treatment.
- This is where the pickling is taking place.

- After pickling, you can see that the sheet.
- It is passing through a bath of zinc chloride flux and then it comes to the molten tin area, and finally it is going to the palm oil, which is going to cover the surface and prevent oxidation of the freshly coated tin.
- Then it goes to the rollers and.
- All the excess tennis report and this is the process of tinning.
- What are the applications?
- It's understood that tin coated steel.
- Why did we talk about this process in the first place?
- Because we know galvanized sheets cannot be used for making containers which store food.
- Make fruits, pickles, etc.
- We know that because it's highly they will be slightly acidic in nature, and they could.

Example or application from the lecture

- So an example for this cathodic metal coating would be coating of tin or nickel metal over iron.
- Let us look at this picture and understand.

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Core explanation · 15:00–19:59

That would provide the right environment for corrosion.

Key details

- And also, if it was zinc, it would have dissolved in this, but here it may we have used tin coating, so it is safe.
- And obviously, that would be the biggest advantage of tinning over galvanization.
- Tin coated steel can be used for making containers to store food, but of course, we know there is a serious disadvantage that there will be intense and accelerated corrosion if.
- Even a small crack is formed, or while coating any gaps left on the surface, this would result in pitting corrosion.
- due to the formation of small anode and large cathode area.
- So in the next class we are going to talk about.
- Another method of protective coating and that is inorganic coating.

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EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

C Revision and retrieval practice